A Scrutiny of the MP-LADS in India: Who Is It For?

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This paper attempts to analyse the pattern and determinants of fund utilisation under the Members of Parliament-Local Area Development Scheme by the Lok Sabha MPs. It indicates that there are political business cycles in spending by MPs. Moreover, it shows that the degree of competition faced by an MP in the last election, his/her age, and political affiliation significantly affect fund utilisation. It also concludes that a higher level of awareness of general citizens and better law and order conditions in states restrict the MPs from misusing funds to gain political mileage.

1 Introduction

In many countries, including capital constrained developing countries like India, elected leaders are not only awarded powers to determine the direction and efficacy of government policy, but also in many cases they are empowered to utilise large amount of public money made available under developmental scheme(s). It is important to examine, therefore, whether elected leaders use such authorities benevolently or make use of that to ensure re-election. In this paper, we attempt to shed light on the basic question of political motivation and to identify some factors behind that from an empirical perspective. For this purpose, we analyse the pattern of utilisation of the Members of Parliament-Local Area Development Scheme (MP-LADS). We also examine the impact of some important factors, such as political party affiliates, age, education, popularity, awareness of citizens, etc, on fund utilisation.

The MP-LADS, a plan scheme fully funded by the central government, was introduced in December 1993. The purpose of this scheme was to delegate authority and empower the MPs to recommend developmental works in their respective constituencies with emphasis on creation of durable community assets based on locally felt needs. Along with the Lok Sabha MPs, the Rajya Sabha MPs are also entitled to take advantage of the MP-LADS in order to respond to locally felt needs.[1]

In 1993-94, when the scheme was launched, an amount of Rs 5 lakh per MP was allotted, which became Rs 1 crore per annum per constituency from 1994-95. This was further increased to Rs 2 crore per annum per MP from 1998-99. Since then, approximately Rs 1,590 crore has been allotted to the MPs every year under the MP-LADS, which is about 0.77% of total plan expenditure of the central government and about 49% of plan expenditure on higher education by the central government in the year 2007-08. A total of Rs 17,625.18 crore has been made available to MPs by the central government under this scheme in the last 14 years. The most striking feature of this scheme is that though the funds under the MP-LADS is released on a half yearly basis, the scheme has the flexibility in the sense that any unspent balance of previous year(s) in non-lapsable. That is, an MP can potentially accumulate Rs 10 crore during five years of his or her tenure and spend that amount just before the next election. Given this backdrop, it is important to study the manner of utilisation of funds under the MP-LADS.

Misuse of authority and control by politicians to gain electoral advantage has been a long-standing phenomenon in many democracies across the world. In his pioneering work, Nordhaus (1975) developed a model of political business cycles, where myopic voters get influenced by pre-election boom caused by expansionary fiscal

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and monetary policies and support the incumbent party with favourable votes. Following Nordhaus (1975), the literature on political business cycles has been enriched by many studies. However, Nordhaus' model came under severe criticism due to its presumption that voters' expectations are formed adaptively. In their seminal paper, Rogoff and Sibert (1988) rehabilitated the political business cycle model in the rational expectations framework. They have shown that in equilibrium, incumbent politicians increase government spending in pre-election periods in an effort to signal competence, resulting in a political business cycle in spending. Considering a similar framework of analysis, Rogoff (1990) has shown that incumbent politicians may strategically manipulate the composition of expenditures in pre-electoral years, by favouring items that are more visible to the electorate. The main prediction that incumbent politicians manipulate the economy to gain electoral advantage has received much empirical support. Shi and Svensson (2002) found that on average, the fiscal deficit increases by 1% of the gross domestic product (GDP) in election years with a much larger effect in developing countries. Alesina et al (1997) found the existence of political fiscal cycles in 13 Organisation for Economic Cooperation and Development (OECD) countries, from 1961 to 1993. Evidence of a political business cycle in spending by local governments was also found by Blais and Nadeau (1992) in Canada, Akhmedov and Zhuravskaya (2004) in Russia, Seitz (2000) in Germany, Petterson-Lidbom (2001) in Sweden, to name a few. This empirical evidence indicates that decentralisation may raise the possibility of a political business cycle in spending, and hence misuse of authority and power by the incumbent politicians at sub-national levels in general. Politicians may tend to misuse their control over funds for their personal benefits. The nexus between local governments and local elites and interest groups might further increase the scale of misuse. Some evidence in favour of this argument has been found in the case of the MP-LADS also (PEO 2001).

Features of the Utilisation Process

Two reports by the Comptroller and Auditor General of India (CAG Report 1993-97 and 1997-2000) have pointed out underutilisation of funds and inadequate implementation of the MP-LADS. Based on a survey of selected authoritative bodies, MPs, and potential beneficiaries of the scheme, covering the period from 1994-95 to 1998-99, Programme Evaluation Organisation (PEO) in 2001 revealed that in more than 25% of the cases, the MPs had not considered local needs while recommending works. It also indicated that local politics seems to have played a significant role in utilisation of the MP-LADS. Similar evidence has been found in case of the MLA-ADS in Madhya Pradesh (Singh et al 2003). PEO (2001) also documented that the MP-LADS suffers from financial mismanagement. Most of the works are done through contractors who enjoy the political patronage of the MP, which is not permitted according to the prescribed guidelines; and the quality of the work done under the MP-LADS is not up to the mark in many cases. These are important findings to understand some features of the utilisation process of the MP-LADS. However, existing studies sidestep the important issue of utilisation of the scheme to gain political mileage, and do not investigate the possible impacts of factors like political affiliation, educational attainment, popularity, and age of MPs, awareness of citizens and law and order conditions in states on fund utilisation. This paper attempts to fill this gap.

In this paper we have focused on the Lok Sabha MPs; since, unlike Rajya Sabha MPs, they are directly elected. Our analysis is based on state-level cumulative and Lok Sabha constituency level data on funds released to Lok Sabha MPs and its expenditure, collected from the Ministry of Statistics and Programme Implementation (MOSPI), which are the best possible data sets available for the purpose of this study.[2]

We find that there is a political business cycle in spending the MP-LADS funds. We finds that MPs accumulate funds by not spending about 47% of the allotted amount during the first half of their term, and spend that accumulated amount together with the newly allotted amount during the second half of their term. This implies that MPs utilise the MP-LADS to ensure their re-election.

The analysis of this paper also demonstrates that younger MPs are less prone to create a political business cycle than older MPs. However, educational attainments of the MPs and their popularity do not have any impact on fund utilisation. Interestingly, it seems that Left MPs spend more symmetrically over their entire term than Centre and Right MPs. The paper also indicates that the degree of awareness of general citizens, law and order conditions and level of infrastructure in states play significant roles in determining the process of fund utilisation. While awareness of general citizens and law and order condition of sates have a positive impact, the level of infrastructure has a negative impact on fund utilisation by the MPs.

The rest of the paper proceeds as follows: in the next section, we investigate whether there is any political business cycle in spending. It also summarises the findings of the econometric analysis of the impact of some factors as discussed before on fund utilisation. Section 3 concludes.

2 MP-LADS and Political Business Cycle

In this section, we examine whether there is any political business cycle in spending the money available under the MP-LADS by the 14th Lok Sabha MPs. This analysis is based on state level (cumulative) and constituency-level data on funds available under this scheme and expenditure figures, collected from the web site of the MOSPI.[3]

Cumulative funds released and expenditure data shows that during the period from December 1993 to October 2006, a total of Rs 11,265.18 crore has been made available under the MP-LADS in India, and Rs 10,009.50 has been spent, i e, 88.85% of the money available under the MP-LADS has been utilised. This cumulative utilisation figure increases to 89.84%, if we consider data till November 2007. There are some variations, though not very high, across states in the cumulative fund utilisation. Out of the 14 major states (Andhra Pradesh, Bihar, Gujarat, Haryana, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Orissa, Punjab, Rajasthan, Tamil Nadu, Uttar Pradesh, and West Bengal) under consideration, which encompass 463 out of 545 MP constituencies, Tamil Nadu ranks the first with 95.64% cumulative utilisation and Bihar ranks the last with 82.98% cumulative utilisation as on

30 November 2007. Table 1 reports the state level cumulative utilisation of funds available under the MP-LADS by the Lok Sabha MPs. On an average, these cumulative utilisation figures appear very encouraging. Based on this observation, one may be tempted to jump to the conclusion that the MPs are performing quite well as far as the utilisation of the MP-LADS funds is concerned. However, from these cumulative figures it is not possible to understand the dynamics of fund utilisation by the MPs over the time span of their term in office, which is very important in drawing such a conclusion.

Table 1: Cumulative Fund Utilisation by Lok Sabha MPs

States	Till 31 October 2006			Till 30 November 2007		
	Amount Available (Rs Crore)	Expenditure Incurred (Rs Crore)	% Spent	Amount Available (Rs Crore)	Expenditure Incurred (Rs Crore)	% Spent
Andhra Pradesh	888.41	800.63	90.12	974.72	890.68	91.38
Bihar	812.78	670.86	82.54	880.59	730.71	82.98
Gujarat	542.53	486.05	89.59	585.59	539.13	92.07
Haryana	212.34	187.43	88.27	235.75	206.43	87.56
Karnataka	577.65	524.34	90.77	628.34	568.58	90.49
Kerala	397.18	356.95	89.87	434.29	391.43	90.13
Madhya Pradesh	611.31	564.05	92.27	667.08	623.35	93.44
Maharashtra	976.91	871.64	89.22	1080.65	971.90	89.94
Orissa	439.74	376.22	85.56	479.65	425.73	88.76
Punjab	272.72	239.56	87.84	301.20	274.07	90.99
Rajasthan	529.73	487.53	92.03	587.44	545.93	92.93
Tamil Nadu	832.60	778.22	93.47	922.48	882.24	95.64
Uttar Pradesh	1,700.16	1,498.28	88.13	1,843.82	1,617.25	87.71
West Bengal	802.73	708.06	88.21	889.59	792.34	89.07
Total of 14 states	9,596.79	8,549.82	89.09	10,511.19	9,459.77	90.00
All India	11,265.18	10,009.50	88.85	12,339.43	11,086.18	89.84

Amount available = Cumulative fund released by the government since the inception in 1993 + interest.
Expenditure incurred = Cumulative expenditure since 1993.
% spent = (Expenditure incurred/Amount available)*100.

As stated before, in 1998-99 the amount allotted per MP in each year under the MP-LADS has been increased to Rs 2 crore, which is released in two equal instalments every year. Moreover, any unspent amount of money is non-lapsable, i e, the MPs can potentially choose the time to recommend works without sacrificing any amount that is made available to them under the scheme. In other words, they can accumulate funds during the initial years, and spend that accumulated amount of money along with the newly allotted amounts in year(s) closer to the next election. Two extreme possibilities are (a) each MP utilises Rs 2 crore in each year, and (b) each MP utilises the entire sum of Rs 10 crore only in the last year of their tenure in office. We can consider the first possibility as the benchmark, which refers to the uniform allocation of funds over time. In contrast, the second possibility may be referred to as the case of "severe political business cycle" in spending, since such behaviour on the part of the MPs can only be justified by the motive of ensuring re-election.

Table 2 reports the fund utilisation calculated from the cumulative figures, by the 14th Lok Sabha MPs. Each of them was entitled to utilise Rs 5 crore during the period from May 2004 to October 2006, i e, during the first two and a half years of their tenure. Data shows that, on an average, taking all 545 MP constituencies together, only Rs 2.7 crore have been spent per MP, leaving more than 46% unutilised, during the first half of their tenure. If we extend the period by another one year (i e, up to November 2007), on an average, the accumulated amount per MP remains as high as Rs 2.3 crore. These figures are more or less same, if we consider each state separately. Out of the 14 major states, in 10 states the average accumulated amount per MP during the first half of their term remained more than Rs 2 crore. There was no decline in the accumulated amount in the following year also, from which the next election was still about one and a half years away.

We get a similar picture from the constituency level detailed data on fund utilisation of 217 MP constituencies belonging to 14 major states, by the 14th Lok Sabha MPs till October 2006. This data has been reported by the respective district authorities to the MOSPI, data of remaining constituencies was not available. We find that on an average, only 45.6% of the allotted amount has been utilised by these 217 MPs, i e, about Rs 2.7 crore out of Rs 5 crore per MP has remained unspent till the end of the first half of their tenure.

On the other hand, we find that the overall cumulative utilisation as on November 2006 is 89%. If we deduct the total expenditure by the 14th Lok Sabha MPs till November 2006 from the cumulative expenditure during the period from December 1993 to November 2006 and divide that net amount by the cumulative funds available till April 2004, we will get the measure of fund utilisation till the end of the 13th Lok Sabha. We take a conservative estimate – that on an average about Rs 2 crore out of a total of Rs 5 crore per MP of the 14th Lok Sabha remained unspent till October 2006. Therefore, the calculated fund utilisation at the end of the 13th Lok Sabha is more than 98% (= 100*[10009.5 – 3*545]/[11265.18 – 5*545]). It indicates that MPs tend to make full use of the total amount of money available during their tenure in office. However, the spending/allocation is not uniform across the years of tenure. The MPs accumulate funds during the first half of their tenure in order to spend more when it is close to the next Lok Sabha election, which is likely to increase the probability of re-election of the existing MPs. We note that we need to have annual fund utilisation data of each Lok Sabha separately to depict

Table 2: Fund Utilisation by the 14th Lok Sabha MPs

States	Number of MPs	As on 31 October 2006		As on 30 November 2007	
		Amount Unspent Per MP (Rs Crore)	% Unspent	Amount Unspent Per MP (Rs Crore)	% Unspent
Andhra Pradesh	42	2.09	41.80	2.00	28.59
Bihar	40	3.55	70.96	3.75	53.53
Gujarat	26	2.17	43.45	1.79	25.53
Haryana	10	2.49	49.82	2.93	41.89
Karnataka	28	1.90	38.08	2.13	30.48
Kerala	20	2.01	40.23	2.14	30.61
Madhya Pradesh	29	1.63	32.59	1.51	21.54
Maharashtra	48	2.19	43.86	2.27	32.37
Orissa	21	3.02	60.50	2.57	36.68
Punjab	13	2.55	51.02	2.09	29.81
Rajasthan	25	1.69	33.77	1.66	23.72
Tamil Nadu	39	1.39	27.89	1.03	14.74
Uttar Pradesh	80	2.52	50.47	2.83	40.46
West Bengal	42	2.25	45.08	2.32	33.08
Total of 14 states	463	2.26	45.23	2.27	32.44
All India	545	2.30	46.08	2.30	32.85

Amount unspent per MP = (Unspent Balance/ Number of MPs),
Unspent balance = Amount available - Expenditure incurred.
% unspent = (Amount unspent per MP/entitled amount of each MP of 14th Lok Sabha)*100.

this feature more clearly. Unfortunately, such data is not available, which induces us to adopt the method as mentioned above. Nonetheless, our analysis offers some evidence of political business cycles in spending by the MPs.

Clearly, from the above findings it seems that the MPs are using the MP-LADS with political motives rather than using it in its true spirit. A strict preference of the MPs for more number of small works over relatively less number of large-scale works of capital nature, of which a large proportion of works are not based on locally felt needs but to please interest groups (PEO 2001), also seems to indicate that the MP-LADS is being misused to gain political mileage. The PEO's study in 2001 found that under the MP-LADS, more that 46% of the works got an allocation of less than Rs 50,000, while works with an allocation of Rs 5 lakh or more constituted only 3.6% of the works. Moreover, the large number of works completed/sanctioned came under the category of roads and bridges. Construction works are generally preferred not only for their noticeable character, but also for the ample opportunities of political reward through the ceremonies of declaration, inauguration, etc (Singh et al 2003).

Determinants of Fund Utilisation

We now turn to examine the possible impact of the personal characteristics of the MPs and of constituency-related factors on fund utilisation. Disaggregated data of 217 Lok Sabha constituencies belonging to 14 major states in India from May 2004 to October 2006, which is the first half of the 14th Lok Sabha MPs' tenure, reveals that there are wide variations in the extent of political business cycles in spending across constituencies.[4] The mean and standard deviation of the ratio of the amount of money utilised to the amount of money made available to the MPs, i e, the fund utilisation, are 0.456 and 0.38, respectively. Only about 10% of the MPs utilised the allocated fund fully, whereas more than 40% of the MPs in the sample utilised less than one-third of the allocated fund during the first half of their tenure. That is, some MPs utilised the fund more or less uniformly over time, whereas others accumulated funds heavily during the initial years of their tenure.

In addition, as expected, there are wide variations among the Lok Sabha MPs in the sample in terms of characteristics, such as political affiliation, popularity, age, education, etc. Also, they represent different constituencies that differ in terms of levels of general awareness of people, law and order conditions, infrastructural facilities, etc. In the sample, most of the MPs (171 out of 217) are more than 45 years old, and the average level of education of the MPs is more than graduation (15.53 years of schooling). None of the MPs, in the sample, has less than eight years of schooling. There are 31 Left, 140 Centre, and 46 Right MPs. Average fund utilisation, during the period of study from May 2004 to October 2006, by the Left, Centre, and Right MPs was 51.7%, 46.1% and 40%, respectively. The data also reveals that there is a wide variation in the extent of the level of competition faced by the MPs in the election, i e, in terms of their popularity. In the sample, the number of contestants per constituency varies from 3 to 32, with the average at 10.5; and the winning margin of votes varies from 0.07% to 61.41%.

The econometric analysis, details of which are relegated to the Appendix (pp 67-68), throws up several interesting insights. First, relatively young MPs are inclined to utilise funds under the MP-LADS more uniformly over time than their elder counterparts. In other words, on an average, the extent of political business cycles in spending increases with an increase in the MP's age. However, surprisingly, we find that formal education of MPs does not have any significant impact on fund utilisation (see Table 3 (p 68) in the Appendix).

Interestingly, political affiliations of the MPs appear to have a significant effect on their attitude towards fund utilisation. The regression results indicate that utilisation of funds under the MP-LADS by the MPs affiliated to Left political parties seems to be more uniform than their Centre and Right counterparts. Thus, Left MPs utilise more than Centre MPs, and Centre MPs utilise more than Right MPs during the initial years of their tenure. Descriptive statistics also show that fund utilisation by Left MPs is 11.7% (5.6%) higher than that by the Right (Centre) MPs during the period from May 2004 to October 2006. In other words, it seems that the political business cycle in spending is most severe in case of Right MPs compared to that in case of Centre and Left MPs, as far as MP-LADS is concerned. The analysis also indicates that a higher level of support base of the MPs in their respective constituencies induces them to act more responsibly and to be less dependent on election strategies that mobilise voters during pre-election period. An alternative interpretation is that a lower degree of competition faced in the last election boosts the MPs to rely less on political business cycles in spending to win in the next election.

It is argued that the extent of effectiveness of any decentralised institutional framework crucially depends on the awareness of citizens and law and order conditions of the state (Bardhan 2002). In case of the MP-LADS also, these factors are likely to play important roles. We find that the awareness of citizens has a positive and significant impact on fund utilisation. A higher level of awareness of people leads to a more uniform utilisation of funds, as desired, over time, and will leave less scope for the MPs to utilise funds with political motives. It supports the argument that a higher level of awareness of citizens will put more pressure on the MPs to function more effectively and transparently. Our results also indicate that poor law and order condition of states negatively affects fund utilisation, leaving more scope for misusing MP-LADS. It implies that a higher level of awareness of citizens and improved law and order conditions in states reduces the extent of political business cycles in spending. We also find that the level of existing infrastructure in states has a negative impact on fund utilisation. A possible explanation of this result is that on an average, states having better infrastructural facilities might lead to less demand for small works, which might induce the MPs more to accumulate funds in the initial years of their term to come up with large-scale works towards the end of their term to impress voters. A careful analysis of the type of works sanctioned by the MPs, on a case-by-case basis, might be helpful to understand the impact of the level of existing infrastructure on the manner of fund utilisation. We leave this issue for future research.

3 Conclusions

We have analysed the manner of utilisation of funds under the MP-LADS by the Lok Sabha MPs and its determinants. An important aspect of our analysis is the focus on political motives of the MPs that leads to political business cycles. The attempt has been to study the political business cycle in spending funds under the MP-LADS, its extent and determinants. We have analysed state level cumulative expenditure and fund availability under the MP-LADS from 1993 to 2006 and from 1993 to 2007, and disaggregated constituency level data of the 14th Lok Sabha MPs during the first half of their tenure, separately.

Our analysis indicates that there are political business cycles in spending by the MPs, the extent of which varies widely across Lok Sabha constituencies. Political affiliation is an important determinant of the extent of a political business cycle. It seems that the extent of political business cycles is lower in case of Left MPs compared to that in case of Centre and Right MPs. We also find that a higher level of support base of the MPs in their respective constituencies, reflected through a lower degree of competition faced in the preceding general election, leads to more uniform fund utilisation over time, and younger MPs are less inclined to create political business cycles. Surprisingly, educational attainments of the MPs do not have a significant impact on fund utilisation. We also find that a higher level of awareness of general citizens and better law and order conditions in states restricts MPs in misusing funds to gain political mileage and that leads to more uniform utilisation of funds over time. These results need to be interpreted with caution, since the sample is restricted due to unavailability of data for all the constituencies. This is a potential limitation of this study. Nonetheless, it provides some interesting insights in understanding the present issue.

The accumulation of funds during the initial period of the MPs' tenure and the resultant political business cycle in spending is made possible primarily due to the non-lapsable character of funds under the MP-LADS. It seems that if unused funds are made lapsable, large-scale misuse of the MP-LADS by incumbent parliamentarians to gain political mileage can be restricted at least to some extent. Other than the aspect of political business cycles in spending, there are several other forms of misuse of the MP-LADS by the MPs. For example, uneven distribution of funds across districts, politically motivated choice of works, poor quality of assets created and their poor maintenance, capture by interest groups and close associates of the MPs, financial mismanagement of the scheme and consequent inflated reporting of the amount spent, etc (PEO 2001). In a nutshell, it appears that the MP-LADS, under which a large amount of money is spent every year, is in existence to help the incumbent MPs to stabilise their membership in the Parliament, rather than to serve its true purpose.

NOTES

1 Lok Sabha members can recommend works for their respective constituencies. Elected members of the Rajya Sabha can recommend works for implementation in one or more districts as they may choose in the state of their election. Nominated members of Lok Sabha (two in the 14th Lok Sabha, out of total 545 MPs) and Rajya Sabha can recommend works for implementation in one or more districts anywhere in the country.

2 Constituency level detailed data is available for the 14th Lok Sabha, and state-level cumulative data has been reported at two time points. Therefore, we do not have strong time series data, which is the most appropriate to examine the trend in fund utilisation. Nonetheless, using information on fund released along with the information on cumulative expenditures, one can draw reasonable inferences about the existence of political business cycles.

3 www.mospi.gov.in

4 The choice of the sample was restricted by the availability of non-cumulative data on constituency-wise fund availability and expenditure incurred, which are essential for this analysis. We note that our sample of observations may not be representative, but this is the best one can get given the unavailability of data. Since the sample consists of 47% of total number of constituencies in 14 major states, the analysis seems to be useful at least to some extent. The 14 major states are Andhra Pradesh, Bihar, Gujarat, Haryana, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Orissa, Punjab, Rajasthan, Tamil Nadu, Uttar Pradesh, and West Bengal.

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Appendix: Econometric Analysis

Variables and Data Sources:

We consider "Fund Utilisation", defined as the ratio of the amount of money utilised (spent or sanctioned for works to be done) to the amount of money made available to the MPs during the period, as the dependent variable. We consider disaggregated data on fund utilisation of 217 Lok Sabha constituencies belonging to 14 major states in India from May 2004 to October 2006. We use the MPs' age as an explanatory variable to control for age-specific factors. We also use years of schooling of the MPs as an explanatory variable to examine the impact of formal education on the manner of fund utilisation. In order to assess the role of political affiliations of the MPs, we use the variable "Political Affiliation". Based on the agenda papers of parties and manifestos released before the 14th Lok Sabha election, we have classified political parties into three groups: Left, Centre, and Right. We have considered CPI, CPI(M), FB, and RSP in the Left group; AGP, BJD, DMK, INC, JD(S), JMM, KEC, MDMK, MNF, NCP, PMK, RJD, SP, TDP, TRS, BSP, LJSP, RPI (A), SAD and SS in the Centre group; and BJP in the Right group. "Political Affiliation" takes the value –1 if the MP belongs to a Left political party, 0 if the MP belongs to a Centre political party, and +1 if the MP belongs to a Right political party.

We consider the variable "Popularity", defined as the ratio of (percentage of total votes received by the MP – percentage of total votes received by the runner-up candidate) to (total number of contestants in that constituency), as a proxy to assess the impact of popularity. The numerator is the margin of votes, as a percentage of total valid votes, with which the MP has defeated his/her immediate competitor. We divide the margin by the total number of contestants to normalise the effects of the number

of contestants, which leads to a comparable measure of popularity. A lower value of "Popularity" also indicates a higher degree of competition faced by a MP in the last Lok Sabha election. The mean and standard deviation of the variable "Popularity" are 1.52 and 2, respectively.

We consider variables "Awareness of People" and "Crimes in States" as proxies of awareness of citizens and law and order conditions, respectively. The variable "Awareness of People" is defined as 100 minus the percentage of out of schoolchildren in the age group 6-14 years in the states in 2005. "Crimes in States" is equivalent to ranks of states on the basis of the rate of total cognisable crimes in the states per thousand population in 2001, its higher value indicates poorer law and order conditions. We have also used the variable "Coverage of Roads in State", defined as road length in kilometres per 100 square km in the year 2000, to control for the impact of the level of existing infrastructure. We note that these three variables are likely to vary across constituencies within the same state. However, the constituency level data on such variables is not available.

Data for this analysis comes from various sources: (a) data related to MP-LADS funds were culled from the web site www.mospi.gov.in, (b) data on age and education of the MPs come from the web site www.loksabha.nic.in, (c) agenda papers of parties and manifestos released before the 14th Lok Sabha election provide information on the political affiliations of MPs, (d) percentage of votes and number of contestants were collected from the Statistical Reports on General Elections 2004, Election Commissions of India, New Delhi, and (e) state level characteristics were collected from the web site www.indiastat.com.

Methodology and Results:

Since the dependent variable "Fund Utilisation" is a fraction, which takes values in the range [0, 1], and we have observations with extreme values of the dependent variable, we postulate the following econometric model, as suggested by Papake and Woolridge (1996).

$$y_i = G(X_i\beta) + u_i, \qquad \ldots(1)$$

where i refers to Lok Sabha constituency. The dependent variable y_i is "Fund Utilisation" in the ith constituency. G is logit link function (cdf), X_i denotes the vector of characteristics of the MPs and their constituencies, β is the vector of coefficients of the explanatory variables and u_i is the error term ($E(u_i/x_i) = 0$). The corresponding log-likelihood function is $l_i(b) = y_i log[G(X_ib)] + (1 - y_i)\ log\ [1 - G(X_ib)]$. The Quasi Maximum Likelihood Estimator (QMLE) of β is obtained from the maximisation problem $\underset{b}{Max} \sum_{i=1}^{n} l_i(b)$.

The QMLE is consistent and $\sqrt{N}$ asymptotically normal irrespective of the actual distribution of y. See Papake and Woolridge (1996) for details of this method. Since the dependent variable is bounded between 0 and 1, the Ordinary Least Squares (OLS) method is inappropriate in this case, because the predicted values from the OLS regression can never be guaranteed to lie in the unit interval. Moreover, the traditional way of transforming the data by a logit function, $log[y/(1-y)]$, is not appropriate in this case, since the model cannot be true for y=1. Therefore, QMLE is the most appropriate in our case. Table 3 reports the results of the Quasi Maximum Likelihood Estimation.

We also estimate the following model using the OLS method to test robustness of our results.

$$y_i = X_i\beta + u_i, \qquad \ldots(2)$$

where the dependent variable is the natural logarithm of "Expenditure", which equals the amount of money utilised (spent or sanctioned for works to be done) during May 2004 to October 2006 by a MP, X_i is the vector of explanatory variable, β is the vector of coefficients, and u_i is the error term ($u_i \sim IID(0, \sigma^2)$). We find similar results (see Table 3).

Table 3: Some Factors of Fund Utilisation

Explanatory Variables	(1) Dependent Variable: Fund Utilisation (QMLE)			(2) Dependent Variable: log (Expenditure) (OLS)	
	Coefficients	ME	p-values	Coefficients	p-values
Age of MP	-0.015	-0.004	0.019	-0.018	0.003
Political affiliation	-0.322	-0.080	0.034	-0.382	0.024
Popularity	0.004	0.001	0.056	0.002	0.062
MP's education	0.020	0.005	0.459	-0.004	0.885
Awareness of people	0.125	0.031	0.000	0.066	0.004
Crimes	-0.027	-0.007	0.075	-0.033	0.012
Coverage of roads	-0.005	-0.001	0.003	-0.005	0.002
Constant	3.212		0.000	18.929	0.000
No of observations	217	217		217	
R-squared		0.249		0.178	
Overall significance test		chi2(7) = 30.90 Prob > chi2 = 0.000		F(7, 208) = 4.64 Prob > F = 0.000	

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